MTH 201 Chapter 1: Application of Integration

1.7 Centers of Gravity

General Idea

Formula (Moments and Center of Gravity/Mass): Suppose $f(x)$ is a continuous nonnegative function. If a plate with uniform density, ρ, occupies the region in the *xy*-plane bounded by the curve given by $y = f(x)$ where $a \le x \le b$ and the *x*-axis, then:

- The moment of the plate about the *y*-axis is

- The moment of the plate about the *x*-axis is

- The center of gravity (centroid) of the plate is $(\bar{x}, \bar{y})$

Example 1: Find the center of gravity of the plate with uniform density which occupies the region:

$R = \left\{(x, y) \middle| -r \leq x \leq r \text{ and } 0 \leq y \leq \sqrt{r^2 - x^2}\right\}$ where r is some positive real number.

Example 2:

a) Sketch the region, R, in the first quadrant bounded by the curves $y=\sqrt{x}$, $y=0$, and $x=4$.

b) Suppose a plate with uniform density occupies the region R. Find the exact center of gravity of the plate.